

Play and Learn — Project Plan

2026-09-22

1. Overview

What we're building: "Play and Learn" — a website with 3 user types:

- **Admin** — create, view, update, delete lessons and quizzes
- **Student** — play games, take quizzes, see their own score
- **Parent** — see their child's scores and progress

Two demo courses go in to prove it works: **Python 101** and **Java 101**.

Stack: HTML, CSS, JavaScript, jQuery, Bootstrap (frontend) · PHP (backend) · MySQL (database)

Team

Name	Role	Owns
Yohannis Alem	Frontend	Game screen, quiz screen, student dashboard · Python 101 content
Goffa Euodia	Backend	Admin CRUD, login for all 3 roles · Java 101 content
Davis Ethan	Database	Schema, score tracking, Admin/Parent progress page

2. 8-Week Schedule

Week	Dates	What we do	Show the professor
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Week	Dates	What we do	Show the professor
1	Sep 17–23	Yohannis: sketch dashboard/game/quiz, set up GitHub + local stack Euodia: set up GitHub + local stack, review sketches Ethan: sketch DB tables, set up GitHub + local stack	Plan, sketches, DB drawing — done

2	Sep 24–30	Yohannis: build login screen HTML/Bootstrap, convert sketches to real HTML/CSS Euodia: login/session logic for all 3 roles, start Admin add-lesson/quiz endpoints Ethan: write + push schema.sql, add seed data	Working login screen
3	Oct 1–7	Yohannis: wire dashboard to real DB fields, build dashboard shell Euodia: finish Admin CRUD (add/view/edit/delete) Ethan: schema fixes as CRUD surfaces issues	Admin page working (video/pics)
4	Oct 8–14	Yohannis: load Python 101 lessons + quizzes Euodia: load Java 101 lessons + quizzes Ethan: verify data integrity as content loads	Two courses loaded and visible
5	Oct 15–21	Yohannis: build matching game + quiz page w/ instant scoring Euodia: build score-saving API endpoints Ethan: build scores table + queries	One working game and one working quiz
6	Oct 22–28	Yohannis: style progress page charts (frontend) Euodia: Admin/Parent feedback message logic Ethan: build progress tracker (scores + time spent)	Progress page with sample scores
7	Oct 29–Nov 4	Yohannis: cross-browser/mobile check, fix UI bugs Euodia: fix backend bugs, security pass on endpoints Ethan: check DB queries for bugs/perf issues	List of bugs fixed

8	Nov 5–10	Yohannis: polish UI, help build slides, practice demo Euodia: write backend/DB run instructions, practice demo Ethan: prep clean sample data for demo, practice demo	Final demo ready to show
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3. Who Does What, Each Week

Every week's team goal splits into three parallel pieces. Work your own column, agree on field/endpoint names up front, then merge via PR.

Week	Yohannis (Frontend)	Euodia (Backend)	Ethan (Database)
1	Sketch dashboard, game, quiz screens. Set up GitHub + local PHP/MySQL.	Set up GitHub + local PHP/MySQL. Review page sketches.	Sketch the DB tables needed. Set up GitHub + local PHP/MySQL.
2	Build login screen (HTML/Bootstrap) first. Turn sketches into real HTML/CSS/Bootstrap w/ placeholder data.	Build login/session logic for all 3 roles. Start Admin addlesson/add-quiz endpoints.	Write and push <code>schema.sql</code> (users, courses, lessons, quizzes, scores) + seed data.
3	Wire dashboard to real DB fields. Build student dashboard shell.	Finish Admin CRUD (add/view/edit/delete lessons + quizzes).	Support backend with any schema fixes CRUD surfaces.
4	Load Python 101 lessons + quizzes into Admin tool.	Load Java 101 lessons + quizzes into Admin tool.	Verify data integrity as both courses load in.
5	Build the matching game and the quiz page w/ instant scoring.	Build the API endpoints the game/quiz pages call to save a score.	Build the scores table + queries score-saving relies on.
6	Style progress page charts (frontend half).	Build Admin/Parent feedback message logic.	Build progress tracker: saved scores + time spent.

7	Cross-browser + mobile check on all frontend pages. Fix UI bugs.	Fix backend bugs found in testing. Basic security pass on endpoints.	Check DB queries for bugs/perf issues under test data.
Week	Yohannis (Frontend)	Euodia (Backend)	Ethan (Database)
8	Polish UI for demo. Help build slides. Practice demo flow.	Write run instructions for the backend/DB setup. Practice demo.	Prep clean sample data for the live demo. Practice demo.

4. Ground Rules

- **Branching:** never push straight to `main`. Branch per feature (e.g. `frontentddashboard`, `backend-login`, `db-schema`), open a PR, get it reviewed, then merge.
- **Don't wait on each other:** build against agreed field names/placeholder data; swap in real data once the DB/backend piece lands.
- **Weekly update to the professor:** every week, before class, post a short update in the group chat — what got finished, what's next — so it can be compiled and sent.
- **We turn in:** a working site (Admin, Student, Parent pages), Python 101 + Java 101 loaded, the database with sample data, simple run instructions, demo slides, and all 8 weekly updates.